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Top Skills

Anthropic Claude
Artificial Intelligence (AI)
Figma (Software)

Baden Morgan

Technical Product Manager (IoT) | Ex-Firmware Engineer | Leading Firmware, Software & App Teams to Deliver Scalable Connected Products

Cape Town, Western Cape, South Africa

Summary

I'm a Technical Product Manager with a strong foundation in firmware and embedded systems, specialising in IoT and connected device ecosystems. Over the past 9+ years at Olarm, I've progressed from embedded engineer to product manager, giving me a unique ability to bridge deep technical execution with product strategy. I work across firmware, software, and mobile teams to define product direction, prioritise roadmaps, and deliver reliable, scalable solutions. A key part of my experience is ensuring that the products we design can be successfully manufactured and deployed at scale. I've worked closely with engineering and production teams to align product development with manufacturing constraints, ensuring a smooth transition from concept to production-ready hardware. In addition to product ownership, I've led software and app development efforts in parallel with hardware development, ensuring cohesive end-to-end product delivery. This includes coordinating cross-functional teams and driving alignment between technical execution, user needs, and business objectives. I'm particularly passionate about building IoT products that integrate hardware, firmware, and software into seamless user experiences—while remaining practical, manufacturable, and scalable. If you're building products in IoT, embedded systems, or connected platforms, I'd love to connect.

Experience

Olarm

9 years 5 months

Technical Product Manager (IoT)

July 2024 - Present (2 years)

City of Cape Town

As a Technical Product Manager, I lead the development of IoT-based products spanning firmware, backend systems, and mobile applications.

I currently manage feature development for the Olarm ONE system, driving integration across device firmware, mobile applications, and supporting platforms including internal support tooling, Home Assistant, and command centre systems used by armed response teams for device management and signal monitoring.

I lead cross-functional teams across firmware, software, and app development, ensuring alignment between engineering execution, product strategy, and customer needs. Leveraging my background in embedded systems, I translate complex technical challenges into clear product direction and prioritised roadmaps.

A key focus of my role is ensuring that products are not only well-designed, but also manufacturable and scalable. I work closely with engineering and production teams to align product development with manufacturing constraints, enabling a smooth transition from concept to production-ready hardware.

In parallel with hardware development, I lead software and mobile app initiatives to deliver cohesive, end-to-end product experiences. This includes coordinating across teams, managing dependencies, and driving delivery from concept through to release.

I am responsible for defining product requirements, prioritising features, and ensuring successful delivery across the full product stack.

Firmware Team Lead

March 2023 - June 2024 (1 year 4 months)

- Led the firmware development team responsible for core IoT device functionality across multiple products
- Drove firmware architecture, development, and release coordination for major product launches, including the Olarm LINK and Olarm MAX
- Managed priorities, technical direction, and delivery timelines for firmware development across a growing product portfolio
- Collaborated closely with hardware, backend, and product teams to deliver integrated, end-to-end system solutions

- Played a key role in scaling firmware systems to support increased product complexity and deployment needs
- Supported international certification efforts for alarm communicators under EN standards, ensuring compliance and readiness for global markets
- Led the setup and scaling of manufacturing capabilities, including defining production processes, selecting equipment, and developing supporting software for assembly and testing

Key Contributions:

- Successfully delivered firmware for the Olarm LINK and Olarm MAX, supporting their release to market
- Contributed to achieving EN certification requirements for alarm communicator products
- Improved alignment between firmware, hardware, and software teams during a period of rapid product and team growth
- Enabled scalable manufacturing and testing processes to support reliable product deployment
- Strengthened firmware architecture and development practices to support long-term product evolution

Embedded Systems Team Lead

January 2022 - February 2023 (1 year 2 months)

- Led the embedded systems team across both hardware and firmware development during a key growth phase
- Managed cross-functional engineering efforts, aligning hardware, firmware, and product development to deliver integrated system solutions
- Oversaw technical direction, prioritisation, and delivery across multiple embedded projects

- Continued development and expansion of the Olarm PRO and PRO 4G platforms, adding new integrations and features to enhance functionality and scalability
- Contributed to early development efforts for next-generation products, including the Olarm LINK and Olarm MAX
- Established and scaled a QA function, introducing structured testing processes to improve product reliability and release quality
- Collaborated closely with product and engineering teams to ensure alignment between technical execution and product goals
- Contributed to the development and scaling of manufacturing processes, ensuring products were designed for reliable production and deployment

Key Contributions:

- Built and integrated QA processes into the development lifecycle, improving product stability and release confidence
- Strengthened coordination between hardware, firmware, and product teams as the organisation scaled
- Supported the evolution of core product platforms while contributing to the development of next-generation devices
- Played a key role in preparing products and systems for manufacturing and wider deployment

Embedded Systems Engineer & Support Team Lead

April 2019 - December 2021 (2 years 9 months)

- Continued hands-on embedded development while building and leading the technical support team, primarily focused on the Olarm PRO and PRO 4G
- Established and scaled the support function, including hiring, training, and mentoring team members, and implementing support systems such as live chat and telephony

- Acted as a key bridge between customers, support, and engineering teams, translating real-world issues into product and system improvements
- Worked directly with customers and installers in live environments to deploy, troubleshoot, and improve system reliability and usability
- Contributed to the development of key embedded functionality, including enabling remote firmware updates over WiFi and GSM on a new microcontroller platform
- Worked across embedded software, system integration, and connectivity to support a reliable and scalable connected device experience

Key Contributions:

- Built and matured the company's technical support capability during a key growth phase
- Improved feedback loops between customers and engineering, driving product and reliability improvements
- Enabled a major step forward in product maintainability and scalability through remote update capability
- Supported manufacturing readiness through production support and field issue resolution

Embedded Systems Engineer

February 2017 - March 2019 (2 years 2 months)

Seapoint, Cape Town

Embedded Systems Engineer – Olarm

- Designed and developed firmware and hardware for IoT-based products, primarily working on the Olarm Paradox and Olarm Universal (now Olarm GEN1)
- Worked across embedded software, circuit design, and system integration to expand product capabilities

- Contributed to communication protocols and integration between devices and backend/mobile platforms
- Collaborated with cross-functional teams to deliver production-ready solutions
- Gained hands-on experience with product installations and customer environments, working directly with customers to deploy and troubleshoot systems
- Collected and fed back real-world customer insights to engineering, contributing to product improvements and increased reliability

Key Contributions:

- Played a key role in early-stage product development and system design for core Olarm devices
- Supported the transition of products from prototype to commercial deployment
- Contributed to early manufacturing setup, ensuring designs aligned with production requirements

Olarm (South Africa)

Intern as an Embedded Software Developer

January 2015 - July 2016 (1 year 7 months)

Unit A108, Woodstock exchange, Cape Town & 76 Regent Rd, Sea Point, Cape Town, 8

I worked part time for Olarm, a small start-up company, and full time during University holidays as an Embedded Software Developer. The main objectives of the job were to develop software that would enable the product being developed to interact with the home alarm systems. This entailed understanding various communications protocols and designing and implementing algorithms for the software. I was also tasked at times to do mechanical design work using Solidworks and assist my employer with circuit design and reviewing said circuits. I gained a lot of experience from this job and got to see what it takes to run a small business.

University of Cape Town

Instructor/Tutor for Intel Galileo course
February 2015 - July 2015 (6 months)
University of Cape Town, White Lab

I was tasked with helping out in a course that would last 1 week per session for 2 sessions with a final day for assessment. I helped structure a course plan and material for the course and helped lecture the material to the people taking part in the course. On the final day people who took part in the course brought their projects they built after doing the course for us to assess. In addition 2 more courses of the same nature were held due to the success of the first.

The last course covered the Intel Edison instead of the Galileo board. I was also tasked with creating lesson plans and learning material for the training sessions where I presented the material and taught the participants how to use the Intel Edison like a microcontroller and more. The topics taught included:

- Basics of the Intel Edison
- GPIO and how to use them
- Analog input
- PWM
- UART communications
- Various sensors and how to use them
- Interacting with the Linux shell
- Setting up and connecting to wifi
- Using the Edison as a basic server
- Basic HTML

Intel
Prototype Engineering Guest
November 2014 - December 2014 (2 months)

I was chosen to design and build a project using Intel's Galileo board for a Demonstration on what it was capable of. I was flown up to Johannesburg to show off my project at a faire hosted by Intel.

Education

University of Cape Town
Bachelor of Science, in Engineering in Mechatronics · (2012 - 2016)

Bishops Diocesan College
High School · (2000 - 2011)